Evaluation of the Discovery Process Through Triveritas Analysis

Structural Compatibility, Natural LEM Ordering, and Self-Updating Properties

Registry: [HOWL-DISC-2-2026]

DOI: 10.5281/zenodo.19528560

Date: March 2026

Domain: Applied Philosophy

Status: Working Methodology

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Section 1 — Purpose

HOWL-DISC-1 describes a discovery process for cross-domain investigation, construction, and verification. The process was formalized from decades of practice and documented with complete transparency including published failures. This paper evaluates that process using an external epistemological framework the process was not designed from.

The framework used is the Triveritas, a three-dimensional evaluation tool developed within the Veriphysics tradition. The Triveritas holds that any claim must survive independent scrutiny on three dimensions before meriting assent. Those three dimensions are:

Logical validity (L). Does the reasoning hold? Is there internal coherence? Does the claim follow from its own premises through principled inference, or is it a collection of assertions patched together after the fact? A claim can be logically beautiful and still wrong — but a claim that is logically incoherent is wrong regardless of what the math or evidence says.

Mathematical coherence (M). Does the quantitative structure work? Are the operations well-defined? Do the mathematical predictions follow from the theory's own principles? A claim can have perfect logic and strong evidence but be quantitatively impossible — and without mathematical verification, that impossibility goes undetected.

Empirical anchoring (E). Has the claim been tested against observation? Has it made predictions that were subsequently confirmed or refuted? A claim can be logically valid and mathematically coherent and describe nothing in the observable world — and without empirical testing, that emptiness goes undetected.

Each dimension catches a class of error the other two cannot catch. High L does not compensate for low M. High M does not compensate for low E. High E does not compensate for low L. The dimensions are independent. A claim must pass all three.

The Triveritas also identifies characteristic failure modes defined by which dimensions pass and which fail. A rationalist trap scores high on L and M but low on E — logically sound, mathematically coherent, empirically untested. A beautiful theory about nothing. An empiricist trap scores high on E but low on L — observed but logically incoherent. Real data with a broken explanation. A scholastic trap scores high on L but low on M and E — logical argument without mathematical rigor or empirical contact. Philosophy that never touches the ground.

This paper asks three questions about DISC-1.

First: is the discovery process structurally compatible with Triveritas evaluation? Do the stages of the process naturally test claims on all three dimensions?

Second: does following the process as described yield results that would pass Triveritas scrutiny? Does the complete pipeline produce outputs that are logically valid, mathematically coherent, and empirically anchored?

Third: what are the consequences of the process's natural ordering of dimensions — logical premise first, empirical evidence second, mathematical verification third — for the kind of omni-domain discovery the process is designed to support?

A clarification at the outset. DISC-1 was not designed from the Triveritas. The process was developed through practice over decades. The compatibility between the two, where it exists, is convergent — arrived at independently through different paths. Where the compatibility breaks down, the gap is identified honestly. Where DISC-1 extends beyond what the Triveritas addresses, the extension is noted without claiming it is validated by the Triveritas framework.

Section 2 — Triveritas Compatibility Mapping

Each stage of the DISC-1 process is examined for where and how strongly it tests claims on each of the three Triveritas dimensions.

Intake

The first stage of DISC-1 filters sources by coherency before admitting claims. The coherency filter checks three properties: internal consistency, measurable results, and tested against reality.

Internal consistency is an L-check. A source that contradicts itself fails the filter regardless of its reputation or the quality of its data.

Measurable results is an M-check. A source that produces no quantifiable outcomes fails the filter regardless of the elegance of its reasoning.

Tested against reality is an E-check. A source that has never been checked against observation fails the filter regardless of its internal consistency or mathematical apparatus.

The intake gate is performing a Triveritas evaluation on the source under different vocabulary. All three dimensions are tested before a claim even enters the system.

L: strong. M: moderate (checks for the existence of measurable outputs, not for mathematical rigor of the outputs themselves). E: strong.

Integration

The second stage holds claims with multi-dimensional metadata — source, temporal context, confidence level, scope, relationships to other holdings. Contradictions are noted and held open rather than prematurely resolved.

Checking logical consistency against existing holdings is an L-operation. Noting where new claims support or contradict existing claims is L-evaluation applied across the entire held body of knowledge.

Checking source metadata and temporal context — "when was this true, under what conditions" — is a partial E-operation. It scopes the empirical validity of the claim rather than treating it as universally true or false.

Mathematical coherence is not explicitly tested at integration. A claim can enter the integration layer, connect beautifully with existing holdings, and proceed without any quantitative check. This is a gap.

L: strong. M: absent. E: moderate (through metadata scoping, not through direct empirical testing).

Evaluation

The third stage determines which claims are worth investigating deeply through three tests.

The materiality test asks "does this change the outcome of something I am building or investigating?" This is an E-question — it checks whether the claim has practical impact on real results.

Scope quantification asks "what percentage of cases does this affect?" This is a weak M-operation — it requires putting a number on the claim's impact, but "what percentage" is a much softer mathematical test than "does the quantitative structure compile."

The fruit-of-the-plant test asks "do the results match the claims?" This is an E-operation — empirical outcome validation.

One-step advancement checks whether each logical step holds before advancing to the next. This is an L-operation — verifying the inferential chain link by link.

Information locality prefers direct observation and primary sources over abstracted consensus. This is an E-preference — favoring high-locality empirical data.

L: strong. M: weak (scope quantification is present but is not rigorous mathematical verification). E: strong.

Construction

The fourth stage builds from known good components and uses compilation as the primary test.

"If it doesn't compile, the understanding is wrong." This is the strongest M-gate in the entire process. The compiler is a mechanical mathematical coherence evaluator. It does not check L (logically incoherent code can compile) and it does not check E (code that compiles can produce wrong outputs). But it checks M absolutely — the quantitative and structural apparatus either works or it does not. Zero ambiguity. Zero human judgment. Zero coherence bias.

Assembling known good components into consistent configurations is an L-operation — compositional logical consistency.

Running the built system and checking whether its outputs match expected behavior is an E-operation — functional empirical testing.

L: strong. M: strong (through mechanical compilation). E: strong (through functional testing).

Verification

The fifth stage tests to find the break rather than confirm the result.

Mechanical verification — running arithmetic through scripts, checking outputs through compilers — is M-verification.

Checking outputs against expected behavior and measured data is E-verification.

The coherence warning — "coherence is not correctness" — is L-awareness applied defensively. The verification layer explicitly recognizes that L alone can deceive, that a logically beautiful framework can mask M and E failures, and that mechanical checking is required precisely because human and LLM coherence evaluation is unreliable for mathematical verification.

L: defensive awareness (used as a warning, not as a positive check). M: strong. E: strong.

Remainder Protocol

The sixth stage separates what failed from what survived when verification finds a break.

This is a meta-operation on the L, M, and E results. It asks: which dimension failed? Which dimensions survived? What does the failure pattern tell us about where the process was vulnerable?

In the documented case study, the framework's L survived (the cross-domain connections were real structural observations), M failed (the arithmetic was wrong), and E's apparent pass was revealed as a coherence artifact (the numbers appeared to match measurements because the framework was internally consistent enough that the evaluators — both human and LLM — could not distinguish coherence from correctness). The remainder protocol correctly identified which dimension failed, preserved the survivors from the undamaged dimensions, and released the failed components.

L: the survivors are correctly identified. M: the specific mathematical failure is documented. E: the distinction between genuine empirical confirmation and coherence-disguised-as-confirmation is explicitly recognized. All three dimensions are addressed at the meta-level.

Compatibility Summary

DISC-1 Stage	L	M	E
Intake	Strong	Moderate	Strong
Integration	Strong	Absent	Moderate
Evaluation	Strong	Weak	Strong
Construction	Strong	Strong	Strong
Verification	Defensive	Strong	Strong
Remainder	Meta-level	Meta-level	Meta-level

The pattern is clear. L is tested at every stage. E is tested at most stages. M is tested strongly only at construction and verification — the final two operational stages before the remainder protocol. The pipeline is L-heavy, E-moderate, and M-late.

Section 3 — Does the Process Yield Triveritas Results?

The question is not whether each stage individually performs a complete Triveritas evaluation. No single stage does. The question is whether the complete pipeline, followed from intake through verification, produces outputs that have been tested on all three dimensions by the time they exit.

The answer is yes. By the time a claim has passed through all six stages, it has been checked for logical coherence (at intake, integration, evaluation, and construction), for mathematical structure (at construction and verification), and for empirical anchoring (at intake, evaluation, construction, and verification). Every dimension is tested. The full pipeline yields Triveritas-compatible results.

The case study demonstrates both the success and the vulnerability. A framework entered the pipeline. It passed intake — the source was the practitioner's own derivation, internally consistent, producing measurable outputs, grounded in observed phenomena. It passed integration — the claims connected with existing holdings across multiple domains, every connection was logically coherent, no contradictions were detected. It passed evaluation — the materiality was maximal (the framework claimed to unify all of physics), the scope was universal, and the fruit appeared to match (the numbers matched measured physical constants). It passed initial construction — code was written, it ran, the outputs matched expectations. It did not pass deep mechanical verification — an arithmetic script checking specific integer calculations against claimed results found errors that 45 days of coherence evaluation by three frontier large language models had not detected.

The pipeline caught the failure. The M-dimension, when finally tested with mechanical rigor at the verification stage, found the break that L-evaluation and apparent E-confirmation had masked. The process worked. The Triveritas filter was applied correctly by the time the full cycle completed.

But 45 days of work accumulated between when M would have caught the error (day one, with a simple script) and when M actually caught it (day forty-six). The pipeline works. The timing of M is the vulnerability.

Section 4 — The LEM Ordering

The DISC-1 process naturally sequences its dimensional testing in a specific order: logical premise first, empirical evidence second, mathematical verification third. This is LEM.

This ordering is not a design choice. It is the order in which information becomes available during cross-domain discovery.

A cross-domain connection always arrives as a logical recognition first. The observation that body coordination timing and game engine frame synchronization share the same structural requirements is a logical recognition — a perception of shared form across domains that do not share vocabulary, tools, or institutional membership. This recognition is L. It is not math. It is not empirical data. It is the perception of structural identity.

Only after the logical recognition can the practitioner ask "is there evidence for this?" The evidence that the body actually coordinates faster than nerves can conduct — measured conduction speeds, measured coordination timing, the observable fact that humans can sprint and catch balls and perform ballet — is the empirical check on the logical observation. This is E. It follows L because you cannot look for evidence of a connection you haven't yet recognized.

Only after the empirical evidence suggests the connection is real can the practitioner formalize it mathematically. The control theory inequality, the carrier elimination table, the activity demand matrix — these are the mathematical verification of a connection that was first recognized logically and then supported empirically. This is M. It follows E because you do not invest the effort of mathematical formalization in connections that the evidence has already rejected.

The ordering is LEM because that is the order in which the investigation becomes possible. You cannot look for evidence of a connection you haven't perceived. You cannot formalize a connection that evidence has rejected. The perception comes first. The evidence comes second. The math comes third.

Section 5 — All Six Orderings

Six possible sequences exist for testing claims across three dimensions. Each has characteristic strengths, weaknesses, and failure modes. Each is valid in its appropriate context. Only one is compatible with omni-domain discovery.

LEM — The explorer's sequence

Start with a logical premise. Find empirical evidence. Verify mathematically. This is DISC-1's natural order.

Strength: maximum discovery breadth. The practitioner can explore widely, find cross-domain connections that no single department would see, and build a large body of structural observations before investing in mathematical formalization. The L and E filters are cheap and fast. The expensive M-filter is applied only to claims that have survived both.

Weakness: late M-catch. Claims can accumulate investment across the L and E stages for extended periods before M identifies a mathematical failure. The CKS experience — 45 days of logically coherent, empirically supported, mathematically broken work — is the characteristic failure.

Omni-domain compatibility: the only compatible ordering. Cross-domain connections are logical recognitions that lack shared mathematical language at the point of discovery. If M is required before E, or before L, the connections are never made because they cannot be formalized until after they have been recognized and empirically checked.

LME — The theoretician's sequence

Start with a logical premise. Formalize mathematically. Then check against observation.

Strength: mathematical rigor before empirical investment. You never build on mathematically broken foundations because M-verification happens before you go looking for evidence.

Weakness: only investigates what can be immediately formalized. Every insight that arrives as a pattern, an observation, or a structural recognition without mathematical language is filtered out before it can develop.

Omni-domain compatibility: incompatible. Cross-domain connections lack shared mathematical language at the point of recognition. Requiring formalization before empirical investigation kills the connections before they can be seen. General Relativity was developed in this sequence, but Einstein was working within a single domain (physics) with an existing mathematical language (differential geometry). The sequence works within a domain. It cannot work across domains.

ELM — The empiricist-scientist's sequence

Start with observation. Build a logical framework. Verify mathematically.

Strength: grounded in reality from the start. The investigation begins from something observed, not something imagined.

Weakness: the logical framework built to explain a single observation may be wrong, and M comes last. You can spend decades with a coherent explanation of a real phenomenon that turns out to be mathematically impossible.

Omni-domain compatibility: partially compatible. Observations can be made in any domain. But this sequence anchors to a single domain's observations and builds explanation from there, rather than recognizing structural identities across domains. The cross-domain recognition is a logical act that this sequence places after the empirical anchor, which means the practitioner is already committed to one domain's data before the cross-domain connection is made.

EML — The data scientist's sequence

Start with observation. Build a mathematical model. Then build a logical explanation.

Strength: data-anchored quantitative models from the start.

Weakness: prediction without understanding. The model fits the data and makes accurate predictions, but nobody knows why. The logical explanation may never arrive.

Omni-domain compatibility: incompatible. Mathematical modeling requires domain-specific data and domain-specific tools. You cannot fit a mathematical model across martial arts timing data and game engine frame synchronization data simultaneously unless you have already recognized — through L — that they are the same problem. The L-recognition must come first. Without it, the data from different domains remains separate and the mathematical models are built independently in separate departments.

MLE — The pure mathematician's sequence

Start with mathematical structure. Build logical interpretation. Then check against observation.

Strength: mathematical rigor from the first step. Everything that follows is built on quantitatively sound foundations.

Weakness: may describe nothing in the observable world. String theory has operated in this sequence for four decades — mathematically rigorous, logically coherent, empirically untested.

Omni-domain compatibility: incompatible. Starting from mathematical structure limits the investigation to connections visible through the mathematical forms already in hand. Every genuinely new cross-domain connection requires a logical recognition that precedes the mathematical formalization. The mathematical language for the connection typically does not exist until after the connection has been recognized and explored. VDR — the arithmetic system the practitioner is building — did not exist as a mathematical object when the logical recognition that "division has three outputs, not one" was first made. The math follows the insight. It cannot precede it.

MEL — The pattern-matcher's sequence

Start with mathematical structure. Look for empirical matches. Then build logical explanation.

Strength: mathematical forms are powerful search tools for finding hidden order in empirical data.

Weakness: mathematical coincidence exists. A mathematical form can match an observation without there being any causal connection. The logical explanation built afterward may be a post-hoc rationalization of a numerical accident.

Omni-domain compatibility: incompatible. Same limitation as MLE — restricted to mathematical forms already available. Cannot discover connections that require new mathematical language.

Summary

Sequence	Strength	Characteristic failure	Omni-domain compatible
LEM	Maximum discovery	Late M-catch (CKS)	Yes — the only one
LME	Rigorous before testing	Misses unformalizable insights	No
ELM	Reality-grounded	Wrong explanations persist	Partial
EML	Data-anchored models	Prediction without understanding	No
MLE	Rigorous from start	Beautiful math about nothing	No
MEL	Powerful pattern search	Post-hoc rationalization	No

Why L must come first

Cross-domain connections live in the logical space between domains. They are not in any single domain's data (E) or any single domain's mathematics (M). They are structural recognitions — perceptions of shared form — that can only be made by a mind operating without domain walls. This is an L-operation. It is the foundational act of omni-domain discovery. If L does not come first, the connections are never seen.

Why E must come second

After L identifies a potential connection, E checks whether the connection corresponds to anything real. "Is there actual evidence that the body coordinates faster than nerves can conduct?" is the reality check on the logical observation. If E fails, the logical connection was wrong and the investigation stops before expensive M-formalization is undertaken. E is the cheap filter that prevents mathematical investment in connections that don't exist.

Why M must come last

Mathematical formalization is the most expensive operation in the pipeline. It requires building tools, writing code, constructing formal frameworks. Deferring M until L and E have both passed means the expensive operation is only performed on claims that are logically coherent and empirically supported. This minimizes wasted mathematical effort. The cost is that M-failures are caught late. The benefit is that the practitioner explores broadly enough to find connections that early M-gating would have prevented.

The practitioner's position

All six orderings are valid in their appropriate contexts. LME is correct for theoretical physics within an established mathematical domain. ELM is correct for empirical science with established observational methods. LEM is correct for omni-domain discovery where the connections lack shared mathematical language at the point of recognition.

The practitioner prefers LEM for maximum concept viability, accepting that it leads to more failures at the M stage. The failures are not wasted. Every failure produces methodology improvements, structural observations that survive the specific failure, and refined tools that feed the next iteration. The lessons learned in a failed discovery process are not lost on the next attempt. They are the foundation of the next attempt.

Section 6 — The Self-Updating Property

The LEM ordering's characteristic failure — late M-catch — is not a permanent vulnerability. It is self-correcting.

The CKS failure was an M-failure caught at the verification stage after 45 days of accumulation. The process responded by updating itself through its own mechanisms.

The verification layer now explicitly prioritizes mechanical verification over coherence evaluation. Before CKS, the practitioner used LLM evaluation as a verification method. After CKS, the practitioner treats LLM evaluation as an exploration tool only and requires compiled arithmetic for verification. The distinction between "LLMs find things" and "compilers check things" is a direct product of the CKS failure, encoded into the process as a permanent methodological update.

The construction layer now uses a programming language with explicit error handling and no hidden approximation. The shift from LLM-assisted derivation to compiled Zig implementation is a direct response to the M-failure: the new tool is incapable of the class of error that killed the old framework.

The integration layer now carries a permanent warning about coherence bias. The observation that three frontier LLMs approved broken arithmetic for 45 days because the framework was internally coherent is indexed as a known failure mode at the integration stage. Every future framework that feels coherent triggers the warning: coherence is not correctness.

The evaluation layer now includes the coherence failure mode in its materiality assessment. When evaluating a new claim, the practitioner explicitly asks whether the claim's apparent strength could be an artifact of coherence rather than independent confirmation. This question did not exist in the evaluation layer before CKS.

These updates were not imposed from outside. They emerged from the process's own remainder protocol. The practitioner identified what failed (M — the arithmetic), documented why the failure was caught late (coherence masked the M-failure during L and E evaluation), identified what survived (structural observations, methodology, the coherence-failure identification itself), and updated the process to address the specific gap.

The evaluation methods built into the process — the materiality test, multi-dimensional indexing, one-step advancement applied to the practitioner's own understanding — are the mechanisms that performed the self-update. The materiality test evaluated the CKS failure as maximally material: it affected the entire framework. The multi-dimensional indexing preserved the structural observations at their appropriate confidence level while releasing the failed arithmetic. The one-step advancement protocol, applied to the practitioner's own methodology, identified the specific gap (LLM coherence evaluation is unreliable for mathematical verification) and advanced exactly one step (use compilers instead).

The process applied itself to its own failure and produced a better version of itself. This is the self-updating property.

The implication for the LEM ordering is significant. The characteristic failure of late M-catch is not permanent. The first major M-failure teaches the process to M-sample earlier. The updated process includes periodic mechanical spot-checks at the integration and evaluation stages — not full mathematical verification, which would destroy exploration breadth, but targeted arithmetic checks on specific claims. "Can you compute this specific result with a script right now?" is a question that takes minutes and would have caught the CKS error on day one.

The second iteration of the process has earlier M-gates than the first. The third iteration will have earlier M-gates than the second, if the second iteration reveals further M-timing vulnerabilities. The process converges toward appropriate M-timing through its own operation. The convergence is not designed. It is emergent from the remainder protocol's systematic identification of where each failure occurred in the pipeline and the subsequent update to address that specific stage.

This self-updating property is itself a Triveritas-testable claim.

L: does the logical structure of the self-update hold? The process identifies a failure mode, traces it to a specific stage, updates that stage to address the failure, and the update logically addresses the identified gap. The inferential chain holds.

M: is the improvement measurable? The post-CKS methodology specifies mechanical verification with compiled arithmetic where the pre-CKS methodology used LLM evaluation. This is a quantifiable change in verification method. Whether it produces measurably fewer M-failures is testable by comparing the error rates of the pre-CKS and post-CKS work.

E: has the update been tested against reality? The VDR specification and the neuroscience paper are the first projects running under the updated methodology. VDR has explicit abandonment conditions — if pi and e cannot be represented in VDR form, the specification documents the boundary and the finding is published. The neuroscience paper's carrier elimination table is mechanically verifiable with a pocket calculator. Both projects will constitute the empirical test of whether the updated methodology catches M-failures earlier than the original. The evidence is accumulating but not yet complete.

Section 7 — Where DISC-1 Extends Beyond Triveritas

The Triveritas evaluates claims. DISC-1 evaluates claims and also evaluates the evaluator. Two areas of DISC-1 address questions the Triveritas does not.

Evaluator reliability

DISC-1's "not holding on" principle is a continuous self-monitoring practice. The practitioner asks, throughout every investigation: am I holding this position because the evidence supports it, or because I have invested effort in it? Am I rejecting this observation because it is wrong, or because it threatens my framework? Am I forcing this resolution because the evidence is sufficient, or because I want to move forward?

These are not L, M, or E questions. They are questions about whether the person performing the L, M, and E evaluation is performing it honestly. The Triveritas assumes an honest evaluator. It provides no mechanism for detecting or correcting evaluator bias. DISC-1 does not assume an honest evaluator. It treats evaluator bias as a known failure mode and builds in continuous self-monitoring as a structural requirement of the process.

This is an extension beyond the Triveritas — a meta-layer that sits above the three-dimensional evaluation and asks whether the evaluation is being performed with integrity. The extension is motivated by direct experience: the CKS failure persisted in part because the practitioner's investment in the framework made it harder to see the M-failure, even though the process's own principles required willingness to kill the framework if verification failed. The "not holding on" principle was already part of the process, but CKS revealed that 45 days of accumulated investment strained even a practiced commitment to non-attachment. The principle is necessary. Maintaining it under pressure requires active, continuous effort.

Evaluator maintenance

DISC-1's physical practice section claims that the cognitive state required for honest evaluation — the ability to hold without gripping, invest without attaching, fail without breaking — degrades without physical training. The process prescribes martial arts practice, proprioceptive training, and interoceptive development not as lifestyle additions but as maintenance requirements for the cognitive substrate that runs the evaluation process.

The claim is specific: proprioceptive awareness (the ability to sense the body's internal state accurately) and interoceptive awareness (the ability to sense internal physiological states like tension, fatigue, and bias accumulation) use the same sensory processing channels that cognitive self-monitoring uses. Training the body to detect internal states trains the mind to detect cognitive bias through the same awareness channel. Physical practice is cognitive maintenance performed through a physical medium.

The Triveritas has no analog for this claim. It evaluates the claim. It does not evaluate the evaluator's physical or cognitive fitness to perform the evaluation.

Whether these extensions are valid is itself Triveritas-testable.

L: does the logical connection between body awareness and cognitive self-monitoring hold? The claim is that both use interoceptive processing and that training one trains the other. The logical structure is coherent — if the sensory channel is shared, training the channel through one modality should improve its function in the other. The inference holds provisionally pending M and E evaluation.

M: can the improvement from physical practice be measured? Proprioceptive acuity is measurable (joint position sense testing, balance platform assessment). Interoceptive accuracy is measurable (heartbeat detection tasks, respiratory load detection). Cognitive bias is measurable (decision-making paradigms that detect sunk-cost effects, confirmation bias, and anchoring). The mathematical structure for testing the claim exists, though the specific experiment connecting physical training to reduced cognitive bias in intellectual work has not been conducted.

E: do practitioners who maintain physical training produce more reliable evaluations than those who do not? This is an empirical question that requires controlled study. The practitioner's own experience is a single data point, insufficient for empirical anchoring beyond the anecdotal level. The claim is stated honestly as a practice-derived principle that has not yet been empirically validated beyond the practitioner's own experience.

The extensions are identified as extensions. They are not claimed to be validated by the Triveritas. They are proposed as questions that the Triveritas framework can evaluate once the necessary empirical research is conducted.

Section 8 — Triveritas Evaluation of DISC-1 Itself

The discovery process is itself a claim: that this method of investigation produces reliable cross-domain results. That claim can be evaluated on all three Triveritas dimensions.

L-evaluation

The process is internally coherent. Each stage follows logically from the previous. The intake filter admits high-coherency sources. The integration layer holds claims with metadata and preserves contradictions. The evaluation layer filters for materiality and advances understanding one step at a time. The construction layer builds from known good components and tests through compilation. The verification layer seeks the break rather than the confirmation. The remainder protocol separates what failed from what survived. The iteration layer feeds every output back as input.

No stage contradicts another. No stage is missing that the process's own logic requires. The falsification criteria are specified and are genuinely capable of killing the process's claims if observed. The process does not exempt itself from its own standards — the practitioner's own major failure is published and documented as evidence that the verification layer operates as described.

The logical structure is complete, coherent, and self-consistent.

L assessment: strong.

M-evaluation

The process is described qualitatively rather than quantitatively in most sections. There is no equation relating intake quality to output quality. There is no formal model of how iteration produces compounding returns. There is no metric for evaluator reliability.

However, the process produces quantitative outputs at the construction and verification stages. Compiled code that runs at measured performance levels. Arithmetic elimination tables verifiable with a calculator. Formal specifications with counted rules and enumerated layers. Published work with DOIs, timestamps, and version histories. The outputs are M-testable even though the process description is not itself a mathematical object.

This gap is real but may not be a defect. A methodology paper describes how to work. It is not itself a mathematical theory. Demanding mathematical structure from a process description is a category expectation: one does not mathematize the method of cooking to validate a recipe. One evaluates the food. The DISC-1 process is evaluated by its outputs, which are M-testable, not by whether the process description itself satisfies M in the way a physics theory would.

M assessment: moderate. The process produces M-testable outputs. The process description itself is qualitative. The moderate score is appropriate for the claim type.

E-evaluation

The process has been tested against reality across multiple domains over years. The body of published work constitutes the empirical record: a game engine and execution system running at measured performance levels, a bootable operating system, a security model, four information theory papers, a physics framework including its complete falsification, a formal arithmetic specification in progress, a neuroscience paper with verifiable elimination tables, a novel, three albums of music, and a methodology for corrected re-investigation after failure.

The published failures constitute the strongest E-evidence. The 390-paper framework invalidated and published with the falsification banner is not an embarrassment in the Triveritas framework — it is the most credible possible evidence that the verification layer operates as described. Anyone can publish successes. Publishing failures under your own name, with complete documentation of what went wrong and why, demonstrates that the process's empirical self-correction is real and not performative.

The work spans multiple domains (engineering, physics, mathematics, neuroscience, music, fiction, philosophy), multiple output types (running software, formal specifications, analytical papers, creative works), and multiple years. The empirical record is broad, deep, and includes the specific type of evidence — published self-correction — that most credibly demonstrates methodological honesty.

E assessment: strong.

Overall assessment

DISC-1 passes L strongly: the logical structure is complete and self-consistent. DISC-1 passes E strongly: the empirical record is extensive and includes published failures. DISC-1 passes M moderately: the process produces M-testable outputs but is not itself a mathematical object, which is appropriate for its claim type.

The process is not a rationalist trap (it has strong E). It is not an empiricist trap (it has strong L). It is not a scholastic trap (it produces measurable outputs). It does not fall cleanly into any of the Triveritas failure modes. Its M-moderate score is a limitation of its claim type, not a failure of its methodology.

Section 9 — Conclusion

The DISC-1 discovery process, evaluated through the Triveritas framework, is structurally compatible with three-dimensional epistemological evaluation across all stages of its pipeline. Following the process as described naturally yields results that have been tested on logical validity, mathematical coherence, and empirical anchoring by the time they exit the verification stage.

The process's natural ordering — logical premise first, empirical evidence second, mathematical verification third — is the only sequencing compatible with omni-domain discovery. Cross-domain connections are logical recognitions that lack shared mathematical language at the point of discovery. Requiring mathematical formalization before empirical investigation or before logical recognition kills the connections before they can be seen. The LEM ordering is not arbitrary. It is necessary for the kind of work the process is designed to do.

The late-M cost of the LEM ordering — the risk of accumulating investment in frameworks that are logically coherent and empirically supported but mathematically broken — is real and is documented through the practitioner's own published failure. The cost is mitigated by the process's self-updating property: the first major M-failure teaches the process to sample M earlier, and the updated process converges toward appropriate M-timing through its own remainder protocol and evaluation methods. The self-correction is not designed from the outside. It emerges from the process operating on its own failures.

DISC-1 extends beyond the Triveritas in two areas: evaluator reliability (continuous self-monitoring for bias) and evaluator maintenance (physical practice as cognitive substrate maintenance). These extensions address an assumption the Triveritas makes but does not test — that the person performing the evaluation is doing so honestly and competently. Whether these extensions are valid is Triveritas-testable and is posed as a question for future empirical research.

The discovery process was not designed from the Triveritas. The compatibility is convergent — arrived at through decades of practice by a different path. The convergence itself is evidence that both frameworks are identifying real structural features of how reliable knowledge is produced, rather than arbitrary features of either framework's construction.

The evaluation is honest because honesty is what the process claims to produce. A dishonest self-evaluation would falsify the process's own claims more decisively than any external critique. The gaps identified — M-absent at integration, M-weak at evaluation, moderate M for the process description itself — are stated plainly because stating them plainly is what the process requires.

The process works. The filter is applied. The ordering is necessary. The failures are published. The methodology updates itself. The rest is the next iteration.

Appendix

A — The Triveritas

The Triveritas is a three-dimensional epistemological evaluation tool developed within the Veriphysics tradition. It holds that any claim must independently satisfy logical validity (L), mathematical coherence (M), and empirical anchoring (E) to merit assent. Each dimension catches a class of error the other two cannot detect. The framework identifies characteristic failure modes by which dimensions fail: rationalist trap (high L and M, low E), empiricist trap (high E, low L), scholastic trap (high L, low M and E). The L, M, and E scales are calibrated with historical anchor points from 0 to 100, providing granular assessment rather than binary pass/fail. The Triveritas was validated internally through three independent proofs (logical, mathematical, empirical) with non-overlapping premise sets, and externally through compatibility analysis across seventeen Western and Eastern epistemological traditions spanning twenty-six centuries and three independent intellectual lineages. Full framework: Veriphysics: The Treatise (external to the Howland Archive).

B — HOWL-DISC-1

The discovery process evaluated in this paper. A systematic method for omni-domain investigation, construction, and verification, formalized from decades of practice. Eight stages: intake, integration, evaluation, construction, verification, remainder protocol, iteration, and cognitive state maintenance. Published with five explicit falsification criteria. DOI: [to be assigned]. Howland Archive, DISC Track.

C — The Six LEM Orderings

Sequence	Description	Omni-domain compatible
LEM	Explorer: premise → evidence → math	Yes
LME	Theoretician: premise → math → evidence	No
ELM	Empiricist: observation → explanation → math	Partial
EML	Data scientist: observation → model → explanation	No
MLE	Pure mathematician: math → interpretation → observation	No
MEL	Pattern matcher: math → data matching → explanation	No

D — CKS Case Study

An attempted unified physics framework deriving all physical constants from hexagonal lattice geometry using exact rational arithmetic. Developed over 45 days with three frontier large language models in adversarial configuration, producing 390 documents across every scientific domain. The models were explicitly instructed to find errors and could not identify any inconsistencies for 45 days. A simple arithmetic verification script falsified the framework in 30 minutes. The failure was an M-failure masked by L-coherence: the framework was internally consistent enough that coherence-based evaluation could not detect the arithmetic errors. The falsification was published with the practitioner's name, full documentation, and a corrected methodology for re-attempting the investigation with mechanical verification replacing coherence evaluation. The CKS experience is the primary evidence for the coherence failure mode identified in this paper and in DISC-1. Zenodo community: Cymatic K-Space Mechanics. Status: invalidated and falsified.

END HOWL-DISC-2-2026

Registry: HOWL-DISC-2-2026

Status: Evaluation paper, DISC Track

Evaluated work: HOWL-DISC-1-2026

Evaluation framework: Triveritas (external to Howland Archive)

Result: Structurally compatible across all three dimensions. L strong. M moderate (appropriate to claim type). E strong. LEM ordering necessary for omni-domain discovery.

Self-updating property mitigates late-M cost. Extensions beyond Triveritas identified and posed for future testing.

The process was evaluated by a framework it was not designed from.

The compatibility is convergent.

The gaps are stated.

The extensions are identified.

The evaluation is honest.

[**HOWL-DISC-2-2026**] Evaluation of the Discovery Process Through Triveritas Analysis. Github: https://github.com/ghowland/papers/tree/main/papers/DISC/HOWL-DISC-2-2026